Finite-difference simulation of apoptotic cell monolayers for Electric Cell-substrate Impedance Spectroscopy

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Abstract

Apoptosis produces a sequence of morphological changes that are reflected in the impedance spectra recorded by electric cell-substrate impedance spectroscopy (ECIS). The analytical Giaever–Keese (GK) model describes confluent, homogeneous monolayers but cannot represent the heterogeneous mixture of live and apoptotic cells that evolves during cell death. A finite-difference (FD) simulation framework previously validated for random cell loss and wound healing is extended here to model apoptosis by representing each cell's membrane shrinkage advancing through three intermediate retraction stages before complete detachment. The model is parameterised against the bare-electrode spectrum reported by Arndt *et al.* [1] for porcine brain capillary endothelial cells (PBCECs) on a $3\,\mathrm{mm}^2$ gold electrode and compared with the observed temporal evolution of impedance magnitude and biophysical parameters during cycloheximide-induced apoptosis.

Introduction

Apoptosis, a regulated form of cell death [2], is a critical determinant of tissue homeostasis and a primary endpoint in drug-efficacy assays. Its onset reshapes adherent cells: the cytoskeleton retracts, tight junctions dissolve, and the basal membrane footprint contracts before the cell body detaches entirely [1]. These changes occur on timescales of minutes to hours and are typically assessed by fluorescent staining or biochemical markers that require endpoint fixation and therefore cannot resolve the progression continuously. Electric cell-substrate impedance spectroscopy (ECIS) provides a non-invasive alternative: a small alternating voltage applied to a microelectrode beneath the culture drives a current whose distribution is modulated by the presence of cells, and the resulting impedance spectrum encodes the collective morphological state of the monolayer [3–5]. The technique was applied to apoptotic monitoring by Arndt *et al.* [1], who showed that the impedance magnitude at 1 kHz decreases with caspase-3 activation, establishing ECIS as a sensitive indicator of apoptosis. The spectral data and the GK biophysical parameters extracted from their publication are reproduced in Fig. 1 and serve as the experimental reference for the present work.

Quantitative interpretation of these spectra requires a biophysical model relating impedance to the structural parameters of the cell monolayer. The Giaever–Keese (GK) ECIS model yields a closed-form expression for the impedance of a microelectrode uniformly covered by a confluent monolayer of identical cylindrical cells [6, 7]. It characterizes the culture through three parameters: the intercellular resistance R_b (tight junctions), the cell-substrate adhesion parameter α, and the membrane capacitance C_m. During apoptosis, R_b collapses within approximately 2 hours of 25 µM cycloheximide (CHX) addition and α declines more slowly over several additional hours [1], while C_m remains temporally invariant and the fraction of live, fully attached cells decreases continuously. No closed-form expression exists for this heterogeneous mixture, as it presents no exploitable symmetry. Here we present a finite-difference (FD) model that explicitly maps the membrane and junction tiles of each individual cell onto a two-dimensional grid, overcoming this limitation. In this representation, partially dead cultures are treated exactly by populating only the live-cell tiles, and the apoptotic state of each cell is encoded through the spatial extent of its membrane. This FD framework is extended here to the apoptotic case, validates the model in the confluent limit against PBCEC data from Arndt *et al.* [1], parameterises the electrode against their reported bare-electrode spectrum, and evaluates the resulting simulated time course against experimental impedance data. The simulation engine, including the finite-difference solver, culture construction routines, and batch sweep scripts, is openly available at `https://github.com/EstebanAcerbo/apoptotic-ecis-sim.git`.

The simulated physical domain is a square region with a centered circular electrode, tiled with square cell stencils. This region is discretized on a uniform grid at subcellular resolution. Each cell occupies a block of $(n+1)^2$ nodes; the $n \times n$ interior nodes carry the membrane admittance, and the border nodes shared with a live neighbor carry the intercellular-junction admittance. Dead cells leave their region as short circuit to the counter electrode, contributing negligible impedance. A circular electrode of area a_e, centered on the domain, is mapped onto the grid; tiles within its radius carry the electrode admittance. Cell size and electrode geometry can be adapted for different ECIS systems. The bare electrode is modelled as a constant-phase element (CPE) with a series constriction resistance, giving the specific electrode impedance. These three parameters were fitted to the bare-electrode spectrum reported by Arndt *et al.*, yielding $A = 3.6 \times 10^{-7}\,\Omega/\mathrm{s}^{n_\mathrm{CPE}}$, $n_\mathrm{CPE} = 0.95$, and $R_s = 260\,\Omega$. Thirteen logarithmically spaced frequencies spanning 1 Hz to 1 MHz are computed per simulation, matching the frequency range of Arndt *et al.*

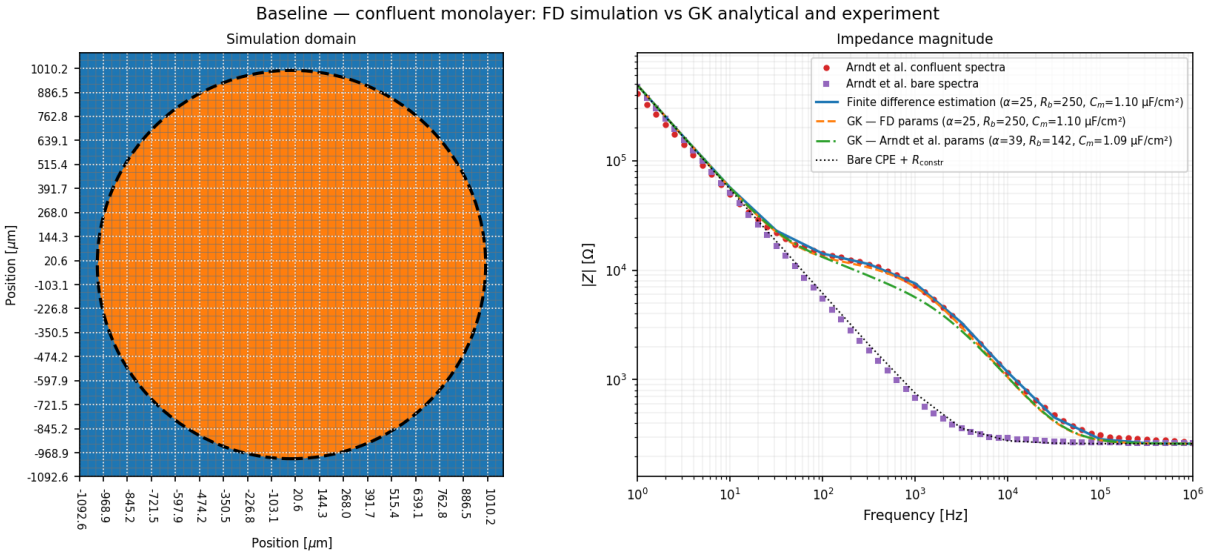


Figure 1: Left: simulated physical domain showing the circular gold electrode (orange) centered on the square insulating substrate, with spatial dimensions in µm. Right: impedance magnitude spectra reported by Arndt *et al.* for a PBCEC confluent monolayer and bare electrode (symbols), together with the CPE fit to the bare-electrode spectrum, with GK analytical fits for the biophysical parameters of Arndt *et al.* and for the parameter set adopted in this work, and the FD simulation result for a confluent monolayer of 1700 µm^2 square cells on a 3 mm^2 gold electrode.

After the cell population distribution is defined, the system's sparse complex matrix of conductances $\mathbf{M}$ is assembled. Off-diagonal elements encode the lateral conductance $c_{\mathrm{lat}} \propto 1/\alpha^2$ between adjacent nodes. Diagonal elements are augmented frequency-by-frequency with the membrane admittance $y_{\mathrm{m}} \propto C_{\mathrm{m}}/2$, the junction admittance $y_{\mathrm{TJ}} \propto 1/Rb$, and the electrode admittance at electrode nodes.

Each voltage node equation equals to the constant elements vector $\mathbf{c}(f)$ defined by electrode and counter electrode potential. The voltage solution $\mathbf{V}$ at each frequency is obtained by solving the linear system $\mathbf{M}(f)\,\mathbf{V}(f) = \mathbf{c}(f)$ and the system impedance is then recovered from the resulting current integral over the electrode.

The apoptotic stages are modelled as three consecutive retractions of the membrane-occupied footprint, with surrounding tiles reverting to short circuit until complete cell detachment, as illustrated in Fig. 2. At each simulation step a fraction P of the total cell population is stochastically recruited into the first apoptotic stage, and cells already in an intermediate stage advance by one level, implementing an irreversible progression through apoptosis. This staging is designed to reproduce the rapid R_{b} collapse and the slower α decline documented by Arndt *et al.*, while C_{m} remains invariant by construction. Each simulation represents a fixed monolayer snapshot; temporal correspondence with the experiment is established a posteriori. To this end, the 1 kHz impedance magnitude at each experimental time point, extracted from *Fig. 6 of Arndt et al.* is matched to the closest precomputed simulation, yielding an estimate of the live and apoptotic cell fractions at each instant.

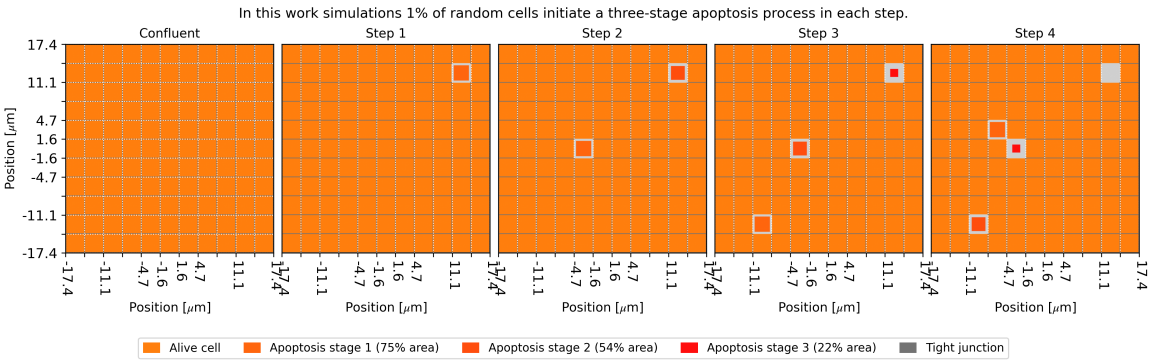
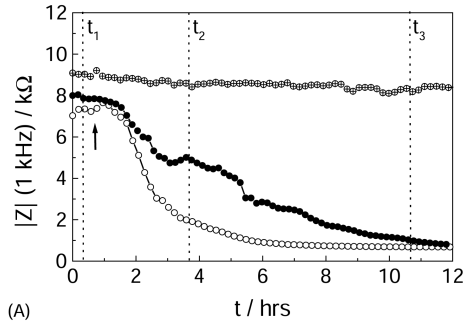


Figure 2: Zoomed monolayer snapshots illustrating the modelled apoptotic progression. From left to right: fully confluent culture, three intermediate stages of progressive membrane retraction, and the final detachment stage. Each cell's colour indicates its apoptotic stage; cells in successive stages occupy a progressively smaller fraction of their original tile area, as indicated in the legend.

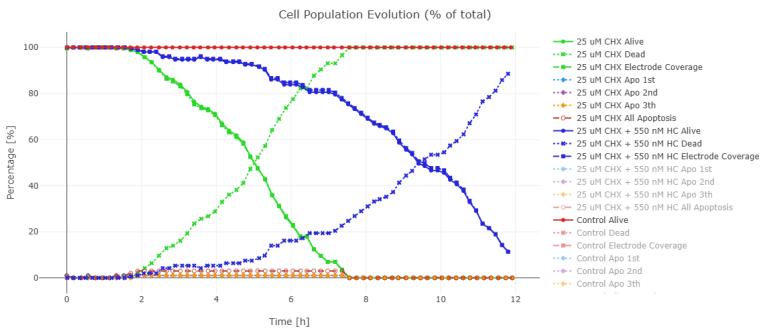
Results

The simulation engine was first validated in the confluent limit against the experimental spectrum and GK analytical fit of Arndt *et al.*, as shown in Fig. 1, where both models are in concordance with the experimental data across the full frequency range. This agreement confirms that the FD framework converges to the GK analytical prediction in the confluent limit. The electrode geometry is that of Arndt *et al.* ($a_e = 3\,\mathrm{mm}^2$); the biophysical parameters adopted here differ from those reported in *Table 2 of Arndt et al.*, as evident from the two GK estimation curves in Fig. 1, yet both parameter sets reproduce the experimental spectrum, indicating a parameter sensitivity comparable to that of the GK analytical expression. The cell area used is $a_c = 1700\,\mu\mathrm{m}^2$ [8]. The experimental temporal reference, extracted from *Fig. 6 of Arndt et al.* [1], is shown in Fig. 3a. After 25 µM CHX addition at $t = 0$, the impedance magnitude at 1 kHz drops monotonically over approximately 6 h (open circles), whereas the 550 nM HC condition (filled circles) retards the decline by approximately a factor of 2.5. A drug-free control (cross-circles) remains stable throughout the 12 h recording.

Using the digitized data, it is possible to estimate the cell population evolution via the closest data-simulation pair, as presented in Fig. 3b. The stochastic apoptosis recruitment model generates an alive-fraction curve that decreases over time as estimated. The scarce repetition of data-simulation pairs indicate that $P = 1\%$ per simulation step is the recruitment fraction consistent with the measured culture state evolution. The drug-free control remains at full confluence throughout. For the CHX-exposed culture, the electrode surface is cleared within approximately 6 h; for the HC-protected condition, full clearance is not achieved within the 12 h assay span. Furthermore, HC supplementation is known to substantially enhance tight-junction resistance [1], so the biophysical parameters calibrated against the drug-free baseline may not fully represent the HC culture; the HC population estimates should therefore be interpreted qualitatively. As the impedance spectra of the HC-exposed culture were not reported in that study, they were not incorporated into the precomputed simulation batch. These timescales are in qualitative agreement with the experimental recordings in Fig. 3a.



(a) Temporal evolution of impedance magnitude at 1 kHz after CHX addition, extracted from Arndt *et al. Fig. 6A* [1]. Open circles: condition without HC. Filled circles: 550 nM HC. Cross-circles: drug-free control. Vertical dashed lines indicate selected time points referenced in the original study.



(b) Simulated population fractions as a function of time for three conditions: 25 µM CHX (green), 25 µM CHX + 550 nM HC (blue), and drug-free control (red). Each condition displays the alive-cell fraction, total dead fraction, and electrode coverage. Individual apoptotic sub-stage fractions is only included for the CHX culture condition for visualization. Time is expressed in hours after CHX addition.

Beyond population fractions, the reconstructed impedance spectra can be analysed within the scope of the GK analytical model by re-fitting GK parameters to each simulated snapshot, following the same procedure applied by Arndt *et al.* [1] to non-confluent culture spectra. Fig. 4 shows the resulting temporal evolution of α, R_b, and C_m for the three simulated conditions, alongside the experimental GK time courses reported by Arndt *et al.* for CHX only exposed culture. The reconstructed R_b captures the rapid initial collapse and the subsequent near-zero values consistent with the dissolution of tight junctions. The reconstructed C_m increases as cells detach from the electrode. Within the GK framework, which assumes full confluency, the reduced membrane coverage produces an apparent increase in the fitted specific capacitance: as fewer cells cover the electrode, the model compensates for the loss of membrane area by inflating C_m rather than explicitly accounting for partial coverage. Contrary to expectation, the reconstructed α initially increases during the apoptotic progression, reverting towards zero only at very low live-cell densities.

This behavior reflects two compounding factors. First, it is not reported by Arndt *et al.* whether the GK parameters were extracted by segmenting the spectrum (fitting α at low frequencies, R_b at intermediate frequencies, and C_m at high frequencies) or by a simultaneous fit over the full frequency range as used in this work, and the choice of strategy can substantially alter the reconstructed trajectories. Second, the GK model is applied here outside its strict domain of validity: as live-cell coverage decreases, the fully-confluent assumption is violated and the reconstructed parameters should be interpreted as effective values rather than literal biophysical quantities. In this context, fitting the GK model to a non-confluent spectrum yields the parameters of the

confluent monolayer whose impedance would be equivalent to the measured one, and should not be interpreted as a direct characterisation of the inhomogeneous culture itself. The finite-difference framework presented here does not share these limitations: by explicitly resolving the impedance contribution of each cell tile and its apoptotic state at the level of individual grid nodes, it provides a physically self-consistent description of spatially inhomogeneous monolayers. The GK re-fitting performed above is therefore presented as a bridge to the existing experimental literature, which reports results in terms of GK parameters, and not as a validation of the GK model under non-confluent conditions.

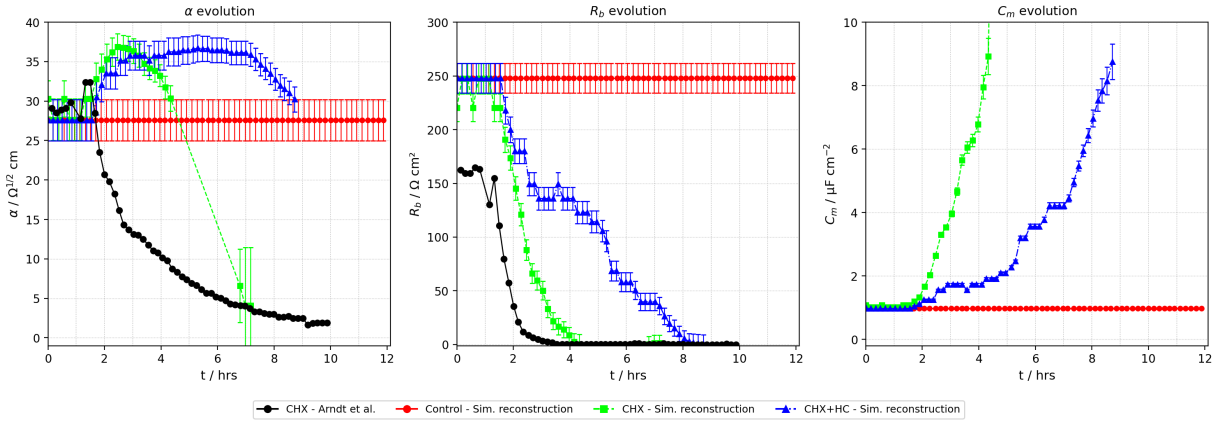


Figure 4: Temporal evolution of the GK biophysical parameters reconstructed from the best-match simulation snapshots: cell-substrate adhesion parameter α (left panel), intercellular resistance R_b (central panel), and membrane capacitance C_m (right panel). Coloured symbols show the three simulated conditions: 25 µM CHX (green squares), 25 µM CHX + 550 nM HC (blue triangles), and drug-free control (red circles). Black filled circles with error bars in the α and R_b panels show the experimental GK fits reported by Arndt *et al.* for the CHX condition.

Conclusions

A finite-difference simulation framework for ECIS is extended to simulate apoptosis through a staged membrane-footprint contraction. The framework accommodates different electrode geometries, cell sizes, and biophysical properties, and the apoptosis progression rate can be tuned through the stochastic recruitment fraction P. In the simulations presented here, cells advance irreversibly through three intermediate morphological stages in which their occupied tile area shrinks to zero before complete detachment, and the resulting impedance evolution reproduces the qualitative difference between fast (CHX) and slow (CHX + 550 nM HC) apoptotic progressions reported by Arndt *et al.* [1], quantifying the fraction of cells in each apoptotic stage at each time point. The simulation parameters were set using the bare and confluent electrode spectra reported in that study. A natural extension of this framework would allow spatially heterogeneous tight-junction disruption by assigning individual R_b values per cell. This would, however, require full impedance spectra at each experimental time point to independently constrain the R_b values for the precomputed simulations.

References

[1] Silke Arndt et al. “Bioelectrical impedance assay to monitor changes in cell shape during apoptosis”. In: *Biosensors and Bioelectronics* 19.6 (2004), pp. 583–594.

[2] John F. R. Kerr. “Shrinkage necrosis: a distinct mode of cellular death”. In: *The Journal of Pathology* 105.1 (1971), pp. 13–20.

[3] Ivar Giaever and Charles R. Keese. “Monitoring fibroblast behavior in tissue culture with an applied electric field”. In: *Proceedings of the National Academy of Sciences* 81.12 (1984), pp. 3761–3764.

[4] Ivar Giaever and Charles R. Keese. “Micromotion of mammalian cells measured electrically”. In: *Proceedings of the National Academy of Sciences* 88.17 (1991), pp. 7896–7900.

[5] Esteban Acerbo, Mariela I. Bellotti, and Fabián J. Bonetto. “Electrical cell-substrate impedance spectroscopy simulation framework for non-confluent cell cultures”. In: *Heliyon* (2024).

[6] Ivar Giaever and Charles R. Keese. “A morphological biosensor for mammalian cells”. In: *Nature* 366.6455 (1993), pp. 591–592.

[7] Chun-Min Lo, Charles R. Keese, and Ivar Giaever. “Impedance analysis of MDCK cells measured by electric cell-substrate impedance sensing”. In: *Biophysical Journal* 69.6 (1995), pp. 2800–2807.

[8] Maj Schneider Thomsen et al. “The blood-brain barrier studied in vitro across species”. In: *PLoS one* 16.3 (2021), e0236770.